

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Final Examination of Semester 2

Subject: 392132 Physics II

Date: 8 March 2014

Year: 2013

Section: 15 - 16

Time: 08.00 – 10.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken inside the examination room.
3. Textbooks are not allowed.
4. Express your answer in English. Other languages will be discarded.
5. A scientific calculator is allowed.
6. Electronic communication device are **NOT** allowed in the examination room.
7. The examination has 6 pages (including this page), 11 questions and a total score of 50 points.
8. Write all your answers on this exam sheet.

วิทยาลัยเทคโนโลยีอุตสาหกรรม

1. Calculate the magnitude of the linear momentum for the following cases: (4 points)

(a) a proton with mass equal to 1.67×10^{-27} kg, moving with a speed of 8.00×10^6 m/s

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(b) a 15.0-g bullet moving with a speed of 300 m/s

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(c) a 66.0-kg sprinter running with a speed of 17.0 m/s

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(d) the Earth (mass = 5.98×10^{24} kg) moving with an orbital speed equal to 2.98×10^4 m/s.

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2. A 0.3 kg volleyball approaches a player horizontally with a speed of 15.0 m/s. The player strikes the ball with her fist and causes the ball to move in the opposite direction with a speed of 20.0 m/s.

(a) What impulse is delivered to the ball by the player? (2 points)

(b) If the player's fist is in contact with the ball for 0.050 0 s, find the magnitude of the average force exerted on the player's fist. (2 points)

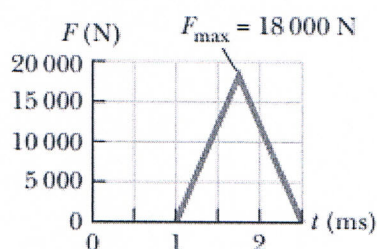
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3. An estimated force vs. time curve for a baseball struck by a bat.

From this curve, determine

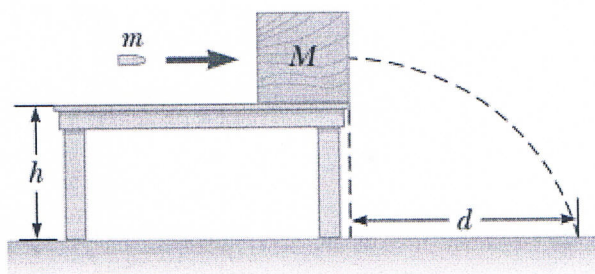
(a) the impulse delivered to the ball. (2 points)

(b) the average force exerted on the ball. (2 points)

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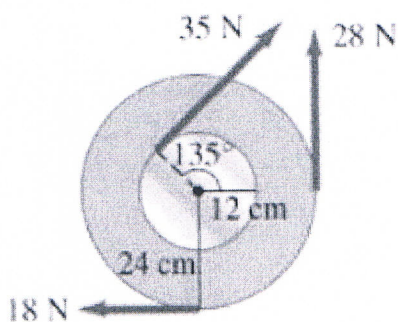
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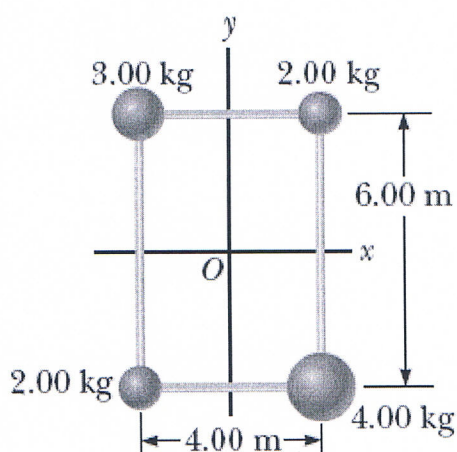


6. A bullet of mass $m = 10.00 \text{ g}$ is fired into a block of mass $M = 3000 \text{ g}$ that is initially at rest at the edge of a table of height $h = 1.20 \text{ m}$. The bullet remains in the block, and after the impact the block lands $d = 2.00 \text{ m}$ from the bottom of the table. Determine the initial speed of the bullet. (8 points)

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7. Calculate the net torque about the axis of the wheel. Assume that a friction torque of $0.4 \text{ N}\cdot\text{m}$ opposes the motion. (3 points)



8. Four objects are held in position at the corners of a rectangle by light rods. (4 points)

- (a) Find the moment of inertia of the system about the x-axis
 (b) find the torque that will produce an angular acceleration of 2.50 rad/s^2 .

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9. A potter's wheel having a radius of 0.50 m and a moment of inertia of $12 \text{ kg} \cdot \text{m}^2$ is rotating freely at 50 rev/min . The potter can stop the wheel in 6.0 s by pressing a wet rag against the rim and exerting a radially inward force of 70 N . Find the effective coefficient of kinetic friction between the wheel and the wet rag. (5 points)

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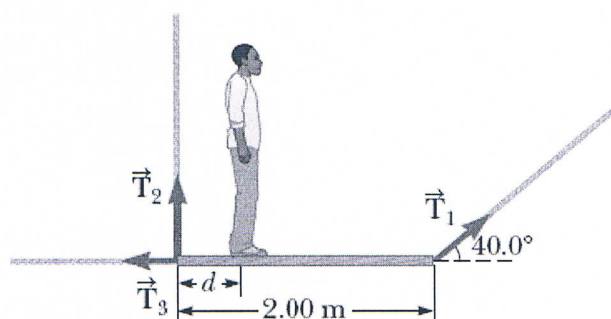
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10. A uniform plank of length 2.00 m and mass 30.0 kg is supported by three ropes, as indicated by the blue vectors. Find the tension in each rope when a 800-N person is $d = 0.500$ m from the left end. (6 points)

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11. A solid, horizontal cylinder of mass 10.0 kg and radius 1.00 m rotates with an angular speed of 7.00 rad/s about a fixed vertical axis through its center. A 0.250-kg piece of putty is dropped vertically onto the cylinder at a point 0.900 m from the center of rotation and sticks to the cylinder. Determine the final angular speed of the system. (3 points)

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Good Luck
Dr. Prakasit Changsuphan